

# SOFTWARE REQUIREMENTS SPECIFICATION

IEEE Std 830-1998 | Realigned Edition

## HunarmandAI

AI-Powered Marketplace for Pakistan's Informal Artisans and Small Businesses

| Field            | Value                   |
|------------------|-------------------------|
| Document Version | 3.0 (Realigned Edition) |
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| Organisation     | FAST-NUCES, Islamabad   |
| Date             | July 2026               |
| Standard         | IEEE Std 830-1998       |
| Status           | Approved Draft          |

### Revision History

| Version | Date     | Author     | Change Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | Mar 2026 | Sana Urooj | Initial draft — all sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.0     | Apr 2026 | Sana Urooj | Full IEEE-conformance review: fixed duplicate use case, corrected use-case IDs, added missing IEEE sections, unified AI timing, added Security NFR, added admin use case, expanded exception flows.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3.0     | Jul 2026 | Sana Urooj | Realigned to confirmed platform requirements and database design: (1) removed gender-specific framing — the platform serves artisans, craftsmen, farmers, and small businesses irrespective of gender; (2) removed the 'Saheli' AI wellness/mental-health module and all related requirements; (3) retained and degendered the Contextual Micro-Learning feature in the Seller panel; (4) replaced the Supabase/PostgreSQL backend with a MongoDB (Mongoose ODM) backend to match the finalised database design; (5) replaced platform-managed third-party courier integration with seller-managed order fulfilment and tracking, since the platform does not handle logistics; (6) broadened the product category taxonomy to the full confirmed category list; (7) changed platform commission from a fixed 5% to an optional, admin-configurable rate; (8) added Credit Card as a payment method alongside JazzCash and EasyPaisa, and removed Cash on Delivery; (9) added the Buyer Credibility Score feature, previously present in the schema but undocumented in the |

|  |  |  |                                                                                                           |
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|  |  |  | SRS; (10) renumbered all use cases, functional requirements, and non-functional requirements accordingly. |
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# 1. Introduction

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## ▪ 1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for HunarmandAI, an AI-powered Progressive Web App marketplace connecting informal artisans, craftsmen, farmers, and small businesses with buyers. It is prepared in conformance with IEEE Std 830-1998 and serves as the authoritative agreement between the development team, stakeholders, and integration partners on exactly what the system must do.

Intended audience: frontend and backend engineers, AI/ML engineers, QA team, academic evaluators at FAST-NUCES, and payment integration partners.

## ▪ 1.2 Document Conventions

- "System" refers to the HunarmandAI platform in its entirety.
- "Seller" refers to the primary users: informal, home-based artisans, craftsmen, farmers, and small business owners, irrespective of gender.
- "Buyer" refers to secondary users: urban consumers, restaurants, and NGO institutional buyers.
- "Admin" refers to platform operations staff with backend management privileges.
- Requirements marked [M] are Mandatory for MVP. Requirements marked [O] are Optional / Phase 2.
- Use Case IDs follow the format UC-NN (two-digit sequential). Functional Requirement IDs follow FR-N-NN (module-sequence). Non-Functional Requirement IDs follow NFR-X-NN.

## ▪ 1.3 Intended Audience & Reading Guide

| Reader                               | Relevant Sections                                                         |
|--------------------------------------|---------------------------------------------------------------------------|
| Frontend / Backend Engineers         | All sections — especially Sections 3, 4, 5, 6, 7                          |
| AI/ML Engineers                      | Sections 2, 3 (UC-03), 4 (FR-2, FR-5), 6 (NFR-P, NFR-S)                   |
| QA / Test Engineers                  | Sections 3, 4, 5 — use cases, functional, and non-functional requirements |
| Academic Evaluators (FAST-NUCES)     | All sections — IEEE conformance verified                                  |
| JazzCash / EasyPaisa / Card Partners | Section 3 (UC-05), Section 4 (FR-4), Section 7.3                          |
| NGO Partners                         | Sections 2.3, 3 (UC-05), 4 (FR-6, FR-7)                                   |

## ▪ 1.4 Project Scope

HunarmandAI aims to turn overlooked home and small-scale production into visible income by connecting informal artisans, craftsmen, farmers, and small businesses in Pakistan — irrespective of gender — with verified buyers. The platform removes traditional e-commerce barriers — digital literacy, language, and mobility — by using AI to automate product listings from a single photograph, and provides contextual micro-learning business-skill modules to help sellers build commercial confidence.

In scope: seller onboarding and AI listing, buyer marketplace, order management, mobile wallet and card payments (JazzCash/EasyPaisha/Credit Card), seller-managed order fulfilment and delivery tracking, contextual micro-learning, admin moderation dashboard, buyer credibility scoring.

Out of scope: native iOS/Android app, international shipping, platform-managed courier/logistics integration (the platform does not manage logistics — sellers are responsible for their own packaging, shipping, and delivery), AI wellness or mental-health features, credit/lending services, social media integration at MVP.

## ▪ 1.5 Definitions, Acronyms, and Abbreviations

| Term / Acronym | Definition                                                                                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| AI             | Artificial Intelligence                                                                                              |
| API            | Application Programming Interface — a defined method by which two software systems communicate                       |
| CLIP           | Contrastive Language–Image Pre-training — OpenAI's vision-language model referenced for product image classification |
| FR             | Functional Requirement — what the system must do                                                                     |
| IEEE           | Institute of Electrical and Electronics Engineers                                                                    |
| JazzCash       | Pakistan's leading mobile wallet service, operated by Mobilink Microfinance Bank                                     |
| EasyPaisha     | Pakistan's first mobile wallet, operated by Telenor Microfinance Bank (now EasyPaisha Digital Bank)                  |
| Mongoose       | An Object Data Modelling (ODM) library that provides schema-based data modelling for MongoDB in Node.js              |
| MongoDB        | A document-oriented NoSQL database used as the platform's primary data store                                         |
| NFR            | Non-Functional Requirement — how well the system must do it                                                          |
| NGO            | Non-Governmental Organisation — institutional buyers and HunarmandAI community partners                              |
| NLP            | Natural Language Processing — AI processing of human language                                                        |
| OTP            | One-Time Password — a temporary numeric code sent via SMS for authentication                                         |
| PECA           | Prevention of Electronic Crimes Act 2016 — Pakistan's primary cybercrime and data law                                |
| PKR            | Pakistani Rupee — the national currency                                                                              |
| PWA            | Progressive Web Application — a web app installable from a browser, requires no app store                            |
| SRS            | Software Requirements Specification — this document                                                                  |
| UC             | Use Case — a description of how an actor interacts with the system to achieve a goal                                 |
| Vercel         | Cloud platform for deploying Next.js applications via serverless functions                                           |

|     |                                                                              |
|-----|------------------------------------------------------------------------------|
| VLM | Vision-Language Model — an AI model that jointly understands images and text |
|-----|------------------------------------------------------------------------------|

▪ 1.6 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2011: Systems and Software Engineering — Life Cycle Processes — Requirements Engineering
- HunarmandAI Business Model Canvas, Sana Urooj, March 2026
- HunarmandAI Industry & Competitive Feasibility Study, Sana Urooj, March 2026
- Anthropic Claude API Documentation, 2026. <https://docs.anthropic.com>
- Next.js 14 Documentation, Vercel Inc., 2025. <https://nextjs.org/docs>
- MongoDB Manual, MongoDB Inc., 2026. <https://www.mongodb.com/docs/manual/>
- Mongoose Documentation, 2026. <https://mongoosejs.com/docs/>
- JazzCash Merchant API Integration Guide, Mobilink Microfinance Bank, 2024
- Pakistan Labour Force Survey 2024-25, Pakistan Bureau of Statistics
- Pakistan Prevention of Electronic Crimes Act (PECA) 2016

## 2. Overall Description

▪ 2.1 Product Perspective

HunarmandAI is a standalone, mobile-first Progressive Web App (PWA) requiring no app store download, ensuring accessibility on low-end smartphones. It operates as a cloud-native platform integrating the following external systems:

- AI/Vision: Anthropic Claude API for photo-to-listing generation (bilingual title, description, category, and price suggestion).
- Backend/Database: Node.js/Express REST API with MongoDB (Mongoose ODM) as the primary data store for all structured data; cloud object storage (CDN-backed) for product images.
- Payments: JazzCash, EasyPaisa, and Credit Card processing for buyer payments in PKR.
- Fulfilment: The platform does not manage logistics. Sellers are responsible for their own packaging, shipping, and delivery; the platform provides order-status tracking tools for both parties.
- Deployment: Vercel (serverless Next.js) for the frontend and API routes; auto-scaling, global CDN.

 **REVISED (v3.0):** Replaced the Supabase/PostgreSQL backend with MongoDB (Mongoose ODM) to match the finalised database design, and replaced platform-managed courier integration (previously Leopards/TCS) with seller-managed fulfilment, since the platform does not handle logistics.

▪ 2.2 System Modules

| Module ID & Name              | Description                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| M1 — Authentication & Profile | OTP-based phone number login; language selection; seller profile setup; wallet linking; session management. |

|                                           |                                                                                                                                                                    |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M2 — AI Product Listing                   | Photo capture → AI Vision → bilingual listing generation (Urdu + English title, description, price, category) in under 10 seconds. Core innovation.                |
| M3 — Seller Storefront                    | Unique shareable URL for each seller. Product grid, ratings, earnings dashboard, WhatsApp contact button.                                                          |
| M4 — Buyer Marketplace                    | Location-filtered browse, full-text Urdu/English search, category filters, product detail, cart, reviews, buyer credibility visibility to sellers.                 |
| M5 — Checkout & Payments                  | End-to-end order: payment via JazzCash/EasyPaisa/Credit Card, optional admin-configurable commission deduction, order confirmation.                                |
| M6 — Seller Order Fulfilment & Tracking   | Seller-managed order lifecycle (Pending → Accepted → Preparing → Shipped → Delivered/Cancelled); buyer-facing status tracking. No third-party courier integration. |
| M7 — Seller Learning Hub (Micro-Learning) | Contextual, bite-sized business-skill lessons triggered by in-app actions (e.g., first listing, underpricing pattern). Gamified with completion badges.            |
| M8 — Admin & Platform Moderation          | Internal dashboard: user management, listing moderation, order oversight, analytics, commission configuration.                                                     |

 **REVISED (v3.0):** Removed the 'Saheli' AI wellness module entirely. Retained and degendered the Contextual Micro-Learning feature as its own module (M7), separate from the removed wellness component. Renamed the former logistics module to reflect seller-managed fulfilment (M6). Renumbered Admin module to M8.

## ▪ 2.3 User Classes and Characteristics

### Sellers (Primary Users)

- Demographics: Informal, home-based artisans, craftsmen, farmers, and small business owners across Pakistan — irrespective of gender — producing jewellery, handicrafts, food items, seeds, produce, fragrances, flowers, clothing, and stone products, among others.
- Characteristics: High craft or production skills; low to zero digital literacy for some sellers; primarily Urdu-speaking; may not read English; some sellers face mobility restrictions and cannot travel to markets or courier offices unaided.
- Representative user: A small-scale artisan in Lahore who hand-embroiders textiles, communicates primarily via WhatsApp voice messages, and has never used an e-commerce platform before.

### Buyers (Secondary Users)

- Demographics: Urban middle-class consumers, restaurants, grocery businesses, and NGO institutional buyers in major Pakistani cities.
- Characteristics: Comfortable with e-commerce; value authentic, locally-sourced, handmade, and small-batch products; seek verified quality assurance and reliable delivery.

### Admins

- Platform operations staff with full system access. Desktop browser users. High technical proficiency.

 **REVISED (v3.0):** Removed the gendered demographic framing ('informal home-based women artisans', representative-user example specifying a woman) and the 'women's economic empowerment' framing under Buyers. The platform serves artisans, craftsmen, farmers, and small

*businesses of any gender. Broadened seller product examples to match the confirmed category list (jewellery, handicrafts, food, seeds, produce, fragrances, flowers, clothing, stones).*

## ▪ 2.4 Operating Environment

- Seller PWA: Android 8.0+ with minimum 2 GB RAM; Chrome 90+ browser; 3G/4G connection (minimum 1 Mbps for photo upload). Offline access to own storefront is supported via service worker caching.
- Buyer Marketplace: Responsive web app; Chrome 90+, Firefox 88+, Safari 14+, Edge 90+; desktop and mobile.
- Server: Next.js API routes and a Node.js/Express REST API layer, deployed as serverless functions on Vercel. Database on MongoDB (Mongoose ODM). Product images stored in CDN-backed cloud object storage.
- AI: All Claude API calls are made server-side from the backend. API keys are never exposed to the client.
- Network: Minimum 3G connection required for photo upload and AI listing. Static pages load on 2G (EDGE) via PWA caching.

## ▪ 2.5 Design and Implementation Constraints

- The platform must be a PWA — no native iOS/Android app is required at MVP stage.
- All seller-facing UI must be available in Urdu (Noto Nastaliq Urdu font) with English as secondary option.
- AI listing generation must complete within 10 seconds for the 90th percentile of requests on 4G.
- Payment processing must support JazzCash, EasyPaisha, and Credit Card at MVP — other international gateways excluded.
- The system must comply with Pakistan's PECA 2016 and relevant branchless-banking regulations for mobile wallet payments.
- The platform does not manage logistics; all packaging, shipping, and delivery is the seller's responsibility.

 **REVISED (v3.0):** *Removed the constraint requiring wellness free-text to be hashed rather than stored, since the wellness feature has been removed. Removed courier-API-specific constraints; added an explicit constraint stating the platform does not manage logistics.*

## ▪ 2.6 Assumptions and Dependencies

- Sellers are assumed to own or have access to a smartphone with a functioning rear camera.
- Anthropic Claude API is assumed to maintain  $\geq 99.5\%$  monthly availability and  $< 5$  second response latency for vision tasks.
- JazzCash and EasyPaisha APIs support merchant-side integration with automatic commission splitting where a commission is configured.
- Sellers have a Pakistani mobile number for OTP-based authentication.
- Sellers are responsible for their own product packaging, shipping, and delivery to buyers; the platform does not provide courier integration at MVP.

- MongoDB Atlas (or equivalent managed MongoDB hosting) supports the platform's expected data volume and query patterns at MVP scale; a paid tier is assumed once usage exceeds free-tier limits.

 **REVISED (v3.0):** Removed the Leopards/TCS courier-API assumption and replaced it with an explicit assumption that sellers self-manage delivery. Replaced the Supabase free-tier assumption with an equivalent MongoDB Atlas assumption.

### 3. Use Cases

Use cases follow the fully-dressed IEEE/Cockburn format: Actor, Priority, Frequency, Preconditions, Trigger, Main Flow, Alternate Flows, Exception Flows, and Postconditions.

| USE CASE 01 — User Registration and OTP Login |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor(s)</b>                               | Primary: Seller or Buyer. Secondary: SMS Gateway.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Priority</b>                               | Mandatory [M]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Frequency</b>                              | Once at registration; recurring at each login.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Preconditions</b>                          | User has a valid Pakistani mobile number and network connectivity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Trigger</b>                                | User opens the app and selects 'Login / Sign Up'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Main Flow</b>                              | <ol style="list-style-type: none"> <li>1. User enters their mobile number.</li> <li>2. System validates the number format (Pakistani, 11-digit) and sends a 6-digit OTP via SMS gateway.</li> <li>3. User enters the received OTP within 5 minutes.</li> <li>4. System verifies the OTP against the stored hashed value and expiry timestamp.</li> <li>5. If this is a new number, system prompts for role selection (Seller / Buyer) and preferred language (Urdu / English).</li> <li>6. System creates the account record and establishes an authenticated session (JWT).</li> <li>7. User is routed to the Seller onboarding flow or Buyer marketplace home, depending on role.</li> </ol> |
| <b>Alternate Flows</b>                        | <p>A1 (Returning user): If the number already exists, System skips role selection and logs the user directly into their existing account.</p> <p>A2 (Resend OTP): If the user does not receive the OTP within 60 seconds, they may request a resend, capped at 3 requests per 10 minutes.</p>                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Exception Flows</b>                        | <p>E1: Invalid mobile number format → System displays an inline validation error and does not send an OTP.</p> <p>E2: Incorrect OTP entered 5 times → System locks OTP entry for that number for 15 minutes.</p> <p>E3: SMS gateway failure → System displays a retry option and logs the failure for admin monitoring.</p>                                                                                                                                                                                                                                                                                                                                                                    |

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| <b>Postconditions</b> | A verified, authenticated session exists for the user.<br>For new sellers, an empty seller profile record is created, pending completion. |
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## USE CASE 02 — Profile and Wallet Management

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor(s)</b>        | Primary: Seller.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Priority</b>        | Mandatory [M]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Frequency</b>       | Occasional — profile setup once, wallet checks frequent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Preconditions</b>   | Seller is authenticated (see UC-01).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Trigger</b>         | Seller navigates to 'My Profile' or 'Wallet'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Main Flow</b>       | <ol style="list-style-type: none"> <li>1. Seller enters or edits: name, shop name, city, product category, shop description, and profile picture.</li> <li>2. Seller optionally adds social media links.</li> <li>3. System uploads the profile picture to cloud object storage and stores the resulting URL.</li> <li>4. System saves the profile and marks it as complete.</li> <li>5. Seller opens the Wallet tab, viewing available balance, pending balance, and transaction history.</li> <li>6. Seller initiates a withdrawal request against the available balance.</li> <li>7. System validates the request and, if eligible, queues the payout to the seller's linked JazzCash/EasyPaisa account.</li> </ol> |
| <b>Alternate Flows</b> | A1 (Edit later): Seller may save a partially complete profile and return to finish it before their first product goes live.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Exception Flows</b> | <p>E1: Withdrawal requested before the seller has completed at least two successful sales → System rejects the request and displays the remaining sales needed before withdrawal is enabled.</p> <p>E2: Requested withdrawal amount exceeds available balance → System rejects the request with an inline error.</p> <p>E3: Image upload failure → System retains the previous profile picture and displays a retry prompt.</p>                                                                                                                                                                                                                                                                                        |
| <b>Postconditions</b>  | <p>Seller profile is persisted and visible on the seller's public storefront (UC-04).</p> <p>Withdrawal requests, once approved, generate a wallet transaction record and update the seller's available and pending balances.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## USE CASE 03 — Generate AI Product Listing

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|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor(s)</b>        | Primary: Seller. Secondary: Claude Vision API.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Priority</b>        | Mandatory [M] — core differentiator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Frequency</b>       | Once per product listed; frequent for active sellers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Preconditions</b>   | Seller is authenticated and has a complete profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Trigger</b>         | Seller taps 'Add Product' and captures or uploads a photo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Main Flow</b>       | <ol style="list-style-type: none"> <li>1. Seller captures a product photo using the phone camera or uploads one from the gallery/webcam.</li> <li>2. System uploads the image to cloud object storage.</li> <li>3. System sends the image to the Claude Vision API with a structured prompt requesting: bilingual title, bilingual description, suggested category, and suggested price range.</li> <li>4. Claude API returns structured JSON within the 10-second performance target.</li> <li>5. System pre-fills the listing form with the AI-generated fields.</li> <li>6. Seller reviews, edits if needed, and confirms the category, price, quantity, and stock status.</li> <li>7. Seller publishes the listing.</li> <li>8. System stores the product record with `approvalStatus: pending`, awaiting admin review (UC-08).</li> </ol> |
| <b>Alternate Flows</b> | <p>A1 (Manual entry): Seller may skip AI generation entirely and manually enter title, description, category, and price.</p> <p>A2 (Multiple images): Seller may add further images to the same listing after the primary photo is processed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Exception Flows</b> | <p>E1: Claude API timeout or failure → System falls back to a blank manual-entry form and notifies the seller that AI suggestions are temporarily unavailable.</p> <p>E2: Image fails content-safety checks → System rejects the image with an explanatory message and does not create a listing.</p> <p>E3: Network interruption during upload → System retries automatically up to 3 times before prompting the seller.</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Postconditions</b>  | <p>A new product record exists with `approvalStatus: pending`.</p> <p>If this is the seller's first published product, the 'Introduction to Selling' micro-learning lesson is queued (see UC-07).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## USE CASE 04 — Inventory and Store Management

|                 |                  |
|-----------------|------------------|
| <b>Actor(s)</b> | Primary: Seller. |
|-----------------|------------------|

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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Priority</b>        | Mandatory [M]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Frequency</b>       | Frequent — daily for active sellers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Preconditions</b>   | Seller has at least one published product.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Trigger</b>         | Seller opens 'My Products' from the storefront dashboard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Main Flow</b>       | <ol style="list-style-type: none"> <li>1. Seller views a list of all their products with current stock status.</li> <li>2. Seller selects a product to edit title, description, price, quantity, or images.</li> <li>3. Seller toggles stock availability between In Stock and Out of Stock.</li> <li>4. System saves changes immediately and reflects them on the buyer-facing marketplace.</li> <li>5. Seller may delete a product, prompting a confirmation dialog ('Are you sure you want to delete this product?').</li> </ol> |
| <b>Alternate Flows</b> | A1 (Bulk stock update): Seller may mark multiple products Out of Stock in a single action, e.g. before travel.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Exception Flows</b> | <p>E1: Seller attempts to delete a product with pending orders → System blocks deletion and instructs the seller to resolve open orders first.</p> <p>E2: Edited product previously approved is now flagged for re-review by admin if the title/description/category changed materially.</p>                                                                                                                                                                                                                                        |
| <b>Postconditions</b>  | Product catalogue reflects the seller's latest edits and stock status in real time.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| USE CASE 05 — Checkout and Payment |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor(s)</b>                    | Primary: Buyer. Secondary: JazzCash / EasyPaisha / Credit Card Payment Gateway.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Priority</b>                    | Mandatory [M]                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Frequency</b>                   | Per order.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>               | Buyer is authenticated and has at least one item in their cart.                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Trigger</b>                     | Buyer taps 'Checkout' from the cart.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Main Flow</b>                   | <ol style="list-style-type: none"> <li>1. Buyer reviews cart items, quantities, and total price.</li> <li>2. Buyer enters or selects a saved delivery address.</li> <li>3. Buyer selects a payment method: JazzCash, EasyPaisha, or Credit Card.</li> <li>4. System redirects to the selected gateway's secure payment flow.</li> <li>5. Buyer completes payment authentication (PIN / OTP / card CVV) on the gateway's interface.</li> </ol> |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>6. Gateway returns a signed payment confirmation callback to the system.</p> <p>7. System verifies the callback signature (HMAC), creates the order record with `payment.status: paid`, and — if a platform commission is configured — deducts the configured percentage before crediting the remainder to the seller's pending wallet balance.</p> <p>8. System notifies the seller of the new order and notifies the buyer of order confirmation with a digital receipt.</p> |
| <b>Alternate Flows</b> | A1 (Cart with items from multiple sellers): System automatically splits the cart into one order per seller at checkout, with a single combined payment charged to the buyer.                                                                                                                                                                                                                                                                                                      |
| <b>Exception Flows</b> | <p>E1: Payment callback signature invalid → System does not mark the order as paid, logs a security event, and instructs the buyer to contact support.</p> <p>E2: Payment declined or cancelled by buyer → Cart is preserved and the buyer is returned to checkout with an error message.</p> <p>E3: Item goes out of stock between cart addition and checkout → System removes the affected item, notifies the buyer, and recalculates the total before payment proceeds.</p>    |
| <b>Postconditions</b>  | <p>Order record(s) exist with status `pending` and payment status `paid`.</p> <p>Seller is notified to begin preparing the order for fulfilment (UC-06).</p>                                                                                                                                                                                                                                                                                                                      |

 **REVISED (v3.0):** Replaces the former 'Doorstep Logistics Fulfilment' use case, which assumed platform-managed courier integration (Leopards/TCS). Per confirmed requirements, HunarmandAI does not manage logistics — the seller is responsible for packaging, shipping, and delivering their own products. This use case instead models the seller-driven order-status lifecycle and buyer-facing tracking that the platform does provide.

| USE CASE 06 — Seller-Managed Order Fulfilment and Delivery Tracking |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor(s)</b>                                                     | Primary: Seller. Secondary: Buyer (receives status updates).                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Priority</b>                                                     | Mandatory [M]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Frequency</b>                                                    | Per order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Preconditions</b>                                                | An order exists with `payment.status: paid` and `status: pending`.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Trigger</b>                                                      | Seller receives a new-order notification (UC-05).                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Main Flow</b>                                                    | <ol style="list-style-type: none"> <li>1. Seller reviews the new order in their order queue and either accepts or rejects it.</li> <li>2. On acceptance, System updates order status to `accepted` and notifies the buyer.</li> <li>3. Seller packages the product and updates status to `preparing`.</li> <li>4. Seller arranges their own delivery or courier of choice and updates status to `shipped`, optionally entering a courier name and tracking reference known only to them.</li> </ol> |

|                        |                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>5. Once the buyer has received the product, seller (or buyer, via confirmation) marks the order `delivered`.</p> <p>6. System moves the corresponding amount from the seller's pending wallet balance to their available balance upon delivery confirmation.</p> <p>7. Buyer is prompted to leave a rating and review for the seller and product.</p> |
| <b>Alternate Flows</b> | <p>A1 (Buyer cancels before acceptance): Buyer may cancel the order any time before the seller accepts it; System refunds the payment and closes the order as `cancelled`.</p> <p>A2 (Seller rejects): Seller may reject an order (e.g. item damaged, unavailable); System refunds the buyer automatically and closes the order as `cancelled`.</p>      |
| <b>Exception Flows</b> | <p>E1: Seller does not update status within a configurable SLA window → System sends a reminder notification to the seller and flags the order for admin visibility.</p> <p>E2: Buyer disputes non-delivery after `shipped` status → Buyer may raise a support ticket (Admin dashboard, UC-08 in the Admin module) for manual dispute resolution.</p>    |
| <b>Postconditions</b>  | <p>Order status history reflects a complete, timestamped audit trail from `pending` through to `delivered` or `cancelled`.</p> <p>Seller's `completedSalesCount` increments on delivery, counting toward the two-sale wallet withdrawal threshold (UC-02).</p>                                                                                           |

 **REVISED (v3.0):** Restores the Contextual Micro-Learning feature as its own module, separate from — and unrelated to — the removed 'Saheli' wellness/mental-health feature. All content has been reviewed to ensure it is gender-neutral.

| USE CASE 07 — Contextual Micro-Learning |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor(s)</b>                         | Primary: Seller.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Priority</b>                         | Optional [O] — Phase 2 enhancement, present at MVP in reduced form.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Frequency</b>                        | Triggered by specific in-app events; a few times per month per active seller.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Preconditions</b>                    | Seller is authenticated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Trigger</b>                          | A qualifying in-app event occurs, e.g. the seller publishes their first product, or a pricing pattern suggests underpricing relative to category averages.                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Main Flow</b>                        | <ol style="list-style-type: none"> <li>1. System detects a qualifying trigger event (e.g. first product published, or a price set materially below the category median).</li> <li>2. System queues a short, contextual lesson card (30–90 seconds, bilingual Urdu/English) relevant to the trigger — e.g. 'Pricing Basics' or 'Writing a Great Product Description'.</li> <li>3. Seller is shown the lesson card the next time they open the app.</li> <li>4. Seller may open the lesson immediately, dismiss it, or select 'Later'.</li> </ol> |

|                        |                                                                                                                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | 5. On completion, System marks the lesson complete and awards a completion badge visible on the seller's profile.                                                                                                             |
| <b>Alternate Flows</b> | A1 (Dismiss): Seller dismisses the lesson without viewing it; System does not re-trigger the same lesson for 30 days.<br>A2 (Later): Seller defers the lesson; it remains available in a 'Learning Hub' tab for later access. |
| <b>Exception Flows</b> | E1: Lesson content fails to load → System logs the failure and silently retries on next app open, without blocking any other seller action.                                                                                   |
| <b>Postconditions</b>  | Lesson completion status is recorded against the seller's profile and reflected in their Learning Hub progress.                                                                                                               |

| USE CASE 08 — Admin Moderates a Product Listing |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor(s)</b>                                 | Primary: Admin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Priority</b>                                 | Mandatory [M]                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Frequency</b>                                | Per new or edited listing pending review.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Preconditions</b>                            | Admin is authenticated with moderation privileges. At least one product has `approvalStatus: pending`.                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Trigger</b>                                  | Admin opens the 'Pending Listings' queue on the moderation dashboard.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Main Flow</b>                                | <ol style="list-style-type: none"> <li>Admin selects a pending listing and reviews the image(s), title, description, category, and price.</li> <li>Admin checks the listing against content policy (no prohibited items, accurate categorisation, no misleading claims).</li> <li>Admin approves the listing, setting `approvalStatus: approved`, making it immediately visible on the buyer marketplace.</li> <li>System notifies the seller that their product was approved.</li> </ol> |
| <b>Alternate Flows</b>                          | A1 (Reject): Admin rejects the listing with a reason code and free-text note; System sets `approvalStatus: rejected` and notifies the seller with the reason, allowing them to edit and resubmit.<br>A2 (Bulk review): Admin may approve multiple straightforward listings in a batch action from the queue view.                                                                                                                                                                         |
| <b>Exception Flows</b>                          | E1: Listing reported by a buyer after approval → Admin may re-review and remove it at any time, notifying the seller of the removal reason.                                                                                                                                                                                                                                                                                                                                               |
| <b>Postconditions</b>                           | The listing's `approvalStatus` is updated and the seller is notified of the outcome.                                                                                                                                                                                                                                                                                                                                                                                                      |

## 4. Functional Requirements

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### ▪ FR-1 — OTP Authentication

**FR-1-01 [M]:** The system SHALL send a 6-digit OTP via SMS to a user-provided Pakistani mobile number upon login/registration request.

**FR-1-02 [M]:** The system SHALL expire each OTP 5 minutes after issuance.

**FR-1-03 [M]:** The system SHALL lock OTP entry for a given number for 15 minutes after 5 consecutive incorrect attempts.

**FR-1-04 [M]:** The system SHALL allow the user to select a preferred display language (Urdu or English) at first registration, and SHALL allow this to be changed later from account settings.

**FR-1-05 [M]:** The system SHALL issue a signed session token (JWT) upon successful OTP verification.

### ▪ FR-2 — AI Image-to-Listing Generation

**FR-2-01 [M]:** The system SHALL accept a product photo captured via device camera or uploaded from gallery/webcam.

**FR-2-02 [M]:** The system SHALL send the uploaded image to the Claude Vision API and SHALL receive a structured response containing a bilingual (Urdu/English) title, bilingual description, suggested category, and suggested price range.

**FR-2-03 [M]:** The system SHALL return AI-generated listing suggestions within 10 seconds for at least 90% of requests under normal network conditions (4G).

**FR-2-04 [M]:** The system SHALL allow the seller to edit any AI-generated field before publishing.

**FR-2-05 [M]:** The system SHALL allow the seller to bypass AI generation entirely and manually enter all listing fields.

**FR-2-06 [M]:** The system SHALL set every newly published listing's `approvalStatus` to `pending` until reviewed by an admin (see FR-8).

### ▪ FR-3 — Order Fulfilment and Tracking

The platform does not manage logistics; sellers arrange their own packaging, shipping, and delivery. The following requirements cover the order-status lifecycle and tracking tools the platform does provide.

**FR-3-01 [M]:** The system SHALL allow the seller to progress an order through a defined status lifecycle: Pending → Accepted → Preparing → Shipped → Delivered, or Cancelled at any point prior to Shipped.

**FR-3-02 [M]:** The system SHALL notify the buyer immediately whenever an order's status changes.

**FR-3-03 [M]:** The system SHALL provide the buyer with a tracking view showing the current status and a full, timestamped status history for each order.

**FR-3-04 [O]:** The system SHALL allow the seller to optionally record a courier name and tracking reference of their own choosing when marking an order as Shipped.

**FR-3-05 [M]:** The system SHALL allow the buyer to cancel an order at any time before the seller accepts it, triggering an automatic refund.

 **REVISED (v3.0):** Renamed from 'Logistics API Integration' and rewritten to remove all references to platform-managed courier booking, webhook-based courier status sync, and automated pickup scheduling with Leopards/TCS. The platform does not manage logistics.

## ▪ FR-4 — Payment Processing, Commission, and Wallet

**FR-4-01 [M]:** The system SHALL integrate with JazzCash, EasyPaisa, and Credit Card payment gateways to process buyer payments in PKR.

**FR-4-02 [M]:** The system SHALL verify the HMAC signature of every payment gateway callback before marking an order as paid.

**FR-4-03 [O]:** The system SHALL support an optional, admin-configurable platform commission percentage; where enabled, the configured percentage SHALL be deducted from each transaction, with the remainder credited to the seller's pending wallet balance.

**FR-4-04 [M]:** The system SHALL only permit a seller to withdraw funds from their wallet after the seller has completed at least two successful (delivered) sales.

**FR-4-05 [M]:** The system SHALL generate a digital receipt for the buyer upon successful payment.

 **REVISED (v3.0):** Removed Cash on Delivery as a payment option — added Credit Card. Changed platform commission from a fixed 5% auto-deduction to an optional, admin-configurable percentage, consistent with the finalised schema's `platformCommissionPercentage` setting. Added the wallet withdrawal threshold requirement (previously undocumented in the SRS despite being a confirmed platform rule).

## ▪ FR-5 — Contextual Micro-Learning

**FR-5-01 [O]:** The system SHALL trigger the 'Introduction to Selling' micro-learning lesson automatically when a seller publishes their first product.

**FR-5-02 [O]:** The system SHALL trigger a 'Pricing Basics' lesson when a seller sets a price materially below the category median for a comparable listing.

**FR-5-03 [O]:** The system SHALL allow the seller to dismiss or defer any triggered lesson without penalty.

**FR-5-04 [O]:** The system SHALL record lesson completion against the seller's profile and display a completion badge.

**FR-5-05 [O]:** The system SHALL make all previously triggered and completed lessons available in a persistent 'Learning Hub' tab.

 **REVISED (v3.0):** Restored this module (previously proposed for removal alongside the unrelated wellness feature). Degendered FR-5-01 ('her first product' → 'their first product'). This module is unrelated to and independent from the removed Saheli wellness/mental-health feature.

## ▪ FR-6 — Marketplace Search and Filter

**FR-6-01 [M]:** The system SHALL support full-text search across product title, description, and tags in both Urdu and English.

**FR-6-02 [M]:** The system SHALL allow buyers to filter marketplace results by category, price range, city, seller rating, and stock availability.

**FR-6-03 [M]:** The system SHALL support the full confirmed product category taxonomy: Jewellery, Handicrafts, Food Items (Preservable), Food Items (Instant Delivery), Seeds, Vegetables, Fruits, Fragrances, Flowers, Clothing (Shawls, Kurtas, Dresses), and Stones.

**FR-6-04 [M]:** The system SHALL allow a buyer to submit a star rating (1–5) and written review for a seller or product, restricted to buyers with a verified completed order for that product.

 **REVISED (v3.0):** Broadened the category filter/taxonomy from the previous narrow set (clothes | food | crafts | agri | beauty | homedecor) to the full confirmed category list, and added an availability filter to match the schema.

## ▪ FR-7 — Buyer Credibility Score

**FR-7-01 [M]:** The system SHALL maintain a credibility score for each buyer, calculated from successful orders, cancelled orders, payment success rate, review history, and account age.

**FR-7-02 [M]:** The system SHALL display a buyer's credibility score to a seller when the seller views buyer information on an incoming order.

**FR-7-03 [M]:** The system SHALL recalculate a buyer's credibility score whenever a relevant event occurs (order completion, order cancellation, payment failure).

 **REVISED (v3.0):** Added — this feature exists in the finalised database schema and original platform requirements but was previously absent from the SRS.

## ▪ FR-8 — Admin Moderation

**FR-8-01 [M]:** The system SHALL queue every newly submitted or materially edited product listing for admin review before it appears on the buyer marketplace.

**FR-8-02 [M]:** The system SHALL allow an admin to approve or reject a pending listing, with a required reason code and optional note on rejection.

**FR-8-03 [M]:** The system SHALL notify the seller of the outcome of admin review.

**FR-8-04 [M]:** The system SHALL allow an admin to suspend or reactivate seller and buyer accounts, and to remove listings reported by users.

# 5. Non-Functional Requirements

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## ▪ NFR-P — Performance

**NFR-P-01 [M]:** The system SHALL support at least 1,000 concurrent users at MVP launch without degradation beyond 2-second median page-load time.

**NFR-P-02 [M]:** AI listing generation SHALL complete within 10 seconds for 90% of requests (see FR-2-03).

**NFR-P-03 [M]:** Marketplace search and filter results SHALL return within 1 second for 95% of queries.

## ▪ NFR-U — Usability

**NFR-U-01 [M]:** The seller-facing UI SHALL be fully available in Urdu using the Noto Nastaliq Urdu font, with English as a secondary language option.

**NFR-U-02 [M]:** Core seller actions (add product, check wallet, view orders) SHALL be reachable within 2 taps from the home screen.

**NFR-U-03 [O]:** The system SHOULD support voice-to-text entry for product descriptions in Urdu for low-literacy sellers.

## ▪ NFR-SC — Scalability

**NFR-SC-01 [M]:** The system SHALL use MongoDB connection pooling to efficiently support the platform's expected concurrent database load.

**NFR-SC-02 [M]:** The backend (Node.js/Express, deployed as serverless functions) SHALL scale horizontally to support growth from MVP to 50,000 active sellers without architectural change.

 **REVISED (v3.0):** Replaced Supabase/PgBouncer-specific scalability wording with MongoDB connection pooling and generic serverless horizontal scaling, consistent with the revised backend architecture.

## ▪ NFR-S — Security

**NFR-S-01 [M]:** All data in transit SHALL be encrypted via TLS 1.2 or higher.

**NFR-S-02 [M]:** OTP codes and password reset tokens SHALL be stored hashed, never in plaintext.

**NFR-S-03 [M]:** The system SHALL implement authorization checks ensuring a seller can only read and modify their own product, order, and wallet records.

**NFR-S-04 [M]:** CNIC numbers SHALL be encrypted at rest and masked (showing only the last 4 digits) in all API responses and UI displays.

**NFR-S-05 [M]:** The system SHALL comply with Pakistan's Prevention of Electronic Crimes Act (PECA) 2016 for data handling and platform conduct.

 **REVISED (v3.0):** Removed the wellness-specific security requirement (hashing of free-text wellness check-in content), since the wellness feature has been removed. Added an explicit CNIC encryption/masking requirement, flagged as a gap during schema review.

## ▪ NFR-D — Data Privacy

**NFR-D-01 [M]:** Personally identifiable information (CNIC, phone number, address) SHALL be encrypted at rest.

**NFR-D-02 [M]:** The system SHALL permanently delete a user's personal data within 30 days of a verified account deletion request, subject to legally required transaction record retention.

 **REVISED (v3.0):** Removed the requirement that raw wellness free-text never be stored, since the wellness feature has been removed in its entirety.

## ▪ NFR-A — Platform Independence

**NFR-A-01 [M]:** The system SHALL be delivered as a Progressive Web App, installable to a device home screen without requiring an app store.

**NFR-A-02 [M]:** The seller's own storefront pages SHALL remain viewable offline via service-worker caching after first load.

# 6. External Interface Requirements

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## ▪ 6.1 User Interfaces

- Mobile-first responsive design, minimum supported viewport 360px width.
- Urdu text rendered in Noto Nastaliq Urdu; English text in a standard sans-serif web font.
- High-contrast, icon-led navigation to support low-literacy users.
- All primary seller actions accessible within 2 taps of the home screen (see NFR-U-02).

## ▪ 6.2 Hardware Interfaces

- Device camera (rear-facing) for product photo capture.
- Device GPS (optional) for auto-filling seller city during onboarding.

## ▪ 6.3 Software Interfaces

| Interface           | Description                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Database            | MongoDB (Mongoose ODM) — primary structured data store for all collections (sellers, buyers, admins, products, orders, etc.). |
|                     |                                                                                                                               |
| Authentication      | OTP-based authentication service issuing signed JWT session tokens.                                                           |
| AI / Vision         | Anthropic Claude API — server-side calls only, for image-to-listing generation.                                               |
| Payment (Primary)   | JazzCash Merchant API — mobile wallet payments.                                                                               |
| Payment (Secondary) | EasyPaisa Merchant API — mobile wallet payments.                                                                              |
| Payment (Tertiary)  | Credit Card processing via PCI-DSS compliant payment gateway.                                                                 |
| SMS Gateway         | Twilio SMS OTP delivery to Pakistani mobile numbers.                                                                          |
|                     |                                                                                                                               |

## ▪ 6.4 Communication Interfaces

- All client-server communication SHALL occur over HTTPS/TLS 1.2+.
- Payment gateway callbacks SHALL be verified via HMAC signature before being trusted.
- SMS delivery for OTP SHALL occur over the SMS gateway's HTTPS API.

# 7. Appendices

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## ▪ Appendix A — Requirements Traceability Matrix

| Use Case                            | Functional Requirements | Non-Functional Requirements |
|-------------------------------------|-------------------------|-----------------------------|
| UC-01 Registration & OTP Login      | FR-1-01 .. FR-1-05      | NFR-S-01, NFR-S-02          |
| UC-02 Profile and Wallet Management | FR-4-04                 | NFR-S-04, NFR-D-01          |

|                                                  |                                    |                    |
|--------------------------------------------------|------------------------------------|--------------------|
| UC-03 Generate AI Product Listing                | FR-2-01 .. FR-2-06                 | NFR-P-02           |
| UC-04 Inventory and Store Management             | FR-6-02, FR-6-03                   | NFR-U-02           |
| UC-05 Checkout and Payment                       | FR-4-01, FR-4-02, FR-4-03, FR-4-05 | NFR-S-01, NFR-P-01 |
| UC-06 Seller-Managed Order Fulfilment & Tracking | FR-3-01 .. FR-3-05                 | NFR-P-01           |
| UC-07 Contextual Micro-Learning                  | FR-5-01 .. FR-5-05                 | NFR-U-01           |
| UC-08 Admin Moderates a Product Listing          | FR-8-01 .. FR-8-04                 | NFR-S-03           |
| (cross-cutting) Buyer Credibility                | FR-7-01 .. FR-7-03                 | NFR-S-03           |
| (cross-cutting) Marketplace Search & Filter      | FR-6-01 .. FR-6-04                 | NFR-P-03           |

## ▪ Appendix B — Data Dictionary (MongoDB Collections)

The data dictionary below reflects the finalised MongoDB (Mongoose) schema and replaces the previous PostgreSQL/Supabase table definitions in full, including removal of the `wellness\_checkins` table and renaming of `lesson\_completions` fields to camelCase.

| Collection               | Key Fields                                                                                                                                                                                          |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sellers                  | phone, cnic (encrypted), shopName, city, productCategory (ref), verificationStatus, wallet.availableBalance, wallet.pendingBalance, wallet.completedSalesCount, wallet.withdrawalEnabled, ratingAvg |
| buyers                   | phone, name, addresses[], accountStatus, credibilityScore.value, credibilityScore.successfulOrders, credibilityScore.cancelledOrders, credibilityScore.paymentSuccessRate                           |
| admins                   | email, passwordHash, role                                                                                                                                                                           |
| categories               | name, parentCategory (ref, self), sortOrder, status                                                                                                                                                 |
| products                 | sellerId (ref), categoryId (ref), title, description, isAiGenerated, images[], price, quantity, stockStatus, approvalStatus, ratingAvg, salesCount                                                  |
| carts                    | buyerId (ref), items[] {productId, sellerId, priceSnapshot, quantity}                                                                                                                               |
| orders                   | orderNumber, buyerId (ref), sellerId (ref), items[] {productId, priceSnapshot, quantity, subtotal}, totalAmount, status, statusHistory[], payment.method, payment.status                            |
| walletTransactions       | sellerId (ref), orderId (ref), type, amount, status, adminApprovedBy (ref)                                                                                                                          |
| conversations / messages | buyerId (ref), sellerId (ref), productId (ref) / conversationId (ref), senderType, text, imageUrl                                                                                                   |

|                                       |                                                                                                                                           |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| reviews                               | buyerId (ref), sellerId (ref), productId (ref), orderId (ref), rating, comment, sellerReply                                               |
| notifications                         | recipientType, recipientId, type, isRead, relatedEntity {type, id}                                                                        |
| reports / supportTickets              | reporterType, reporterId, targetType, targetId, status / raisedByType, raisedById, responses[]                                            |
| lessonCompletions<br>(Micro-Learning) | sellerId (ref), lessonId, triggerType, status (queued   viewed   completed   dismissed), completedAt                                      |
| platformSettings                      | walletSettings.requiredSuccessfulSalesForWithdrawal,<br>platformCommissionPercentage, paymentMethods {creditCard, jazzCash,<br>easyPaisa} |